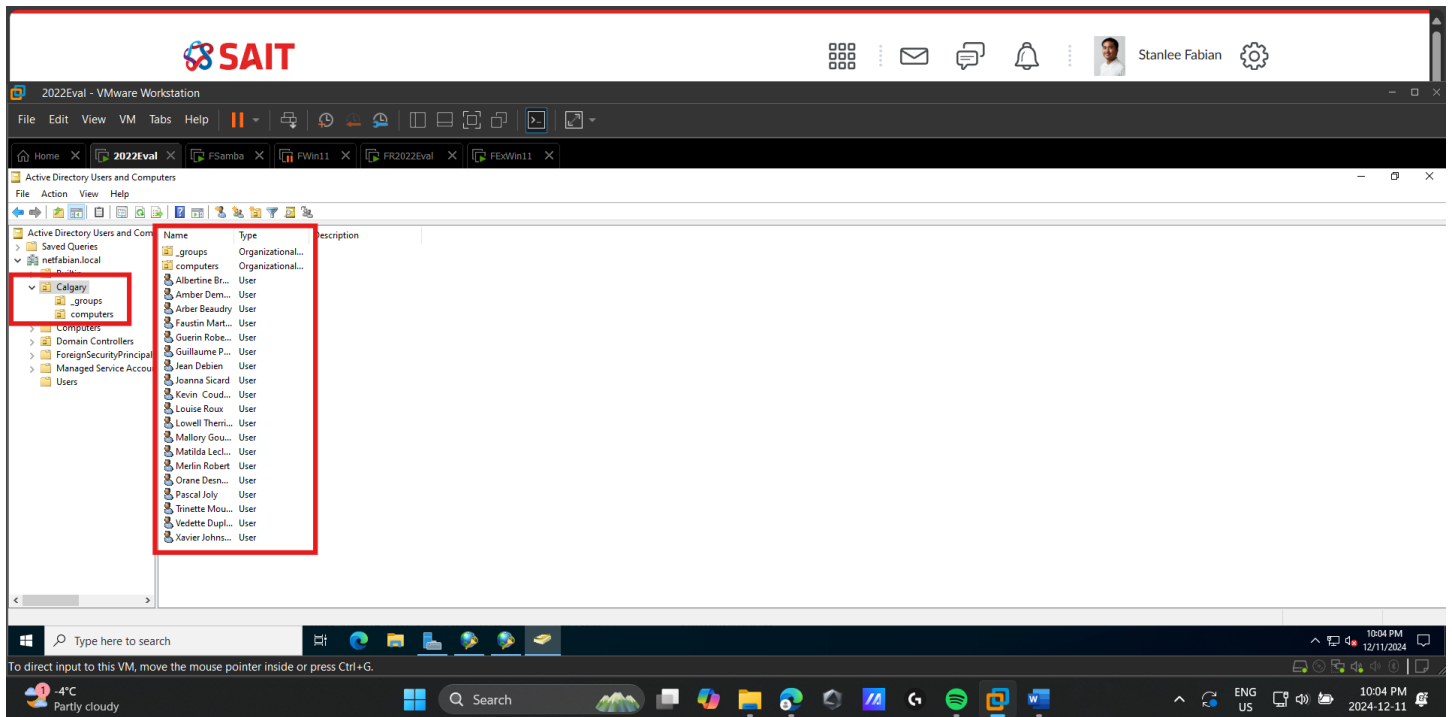
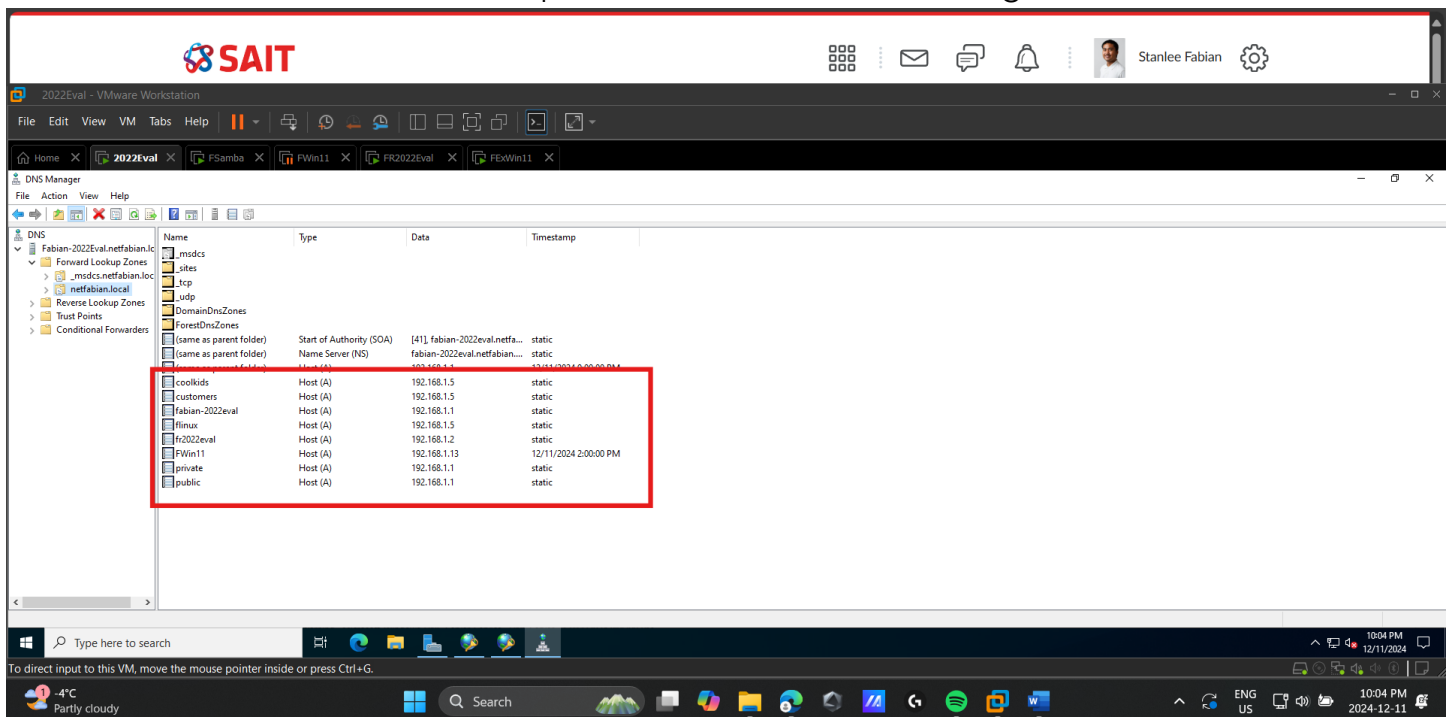


Windows Domain Controller server

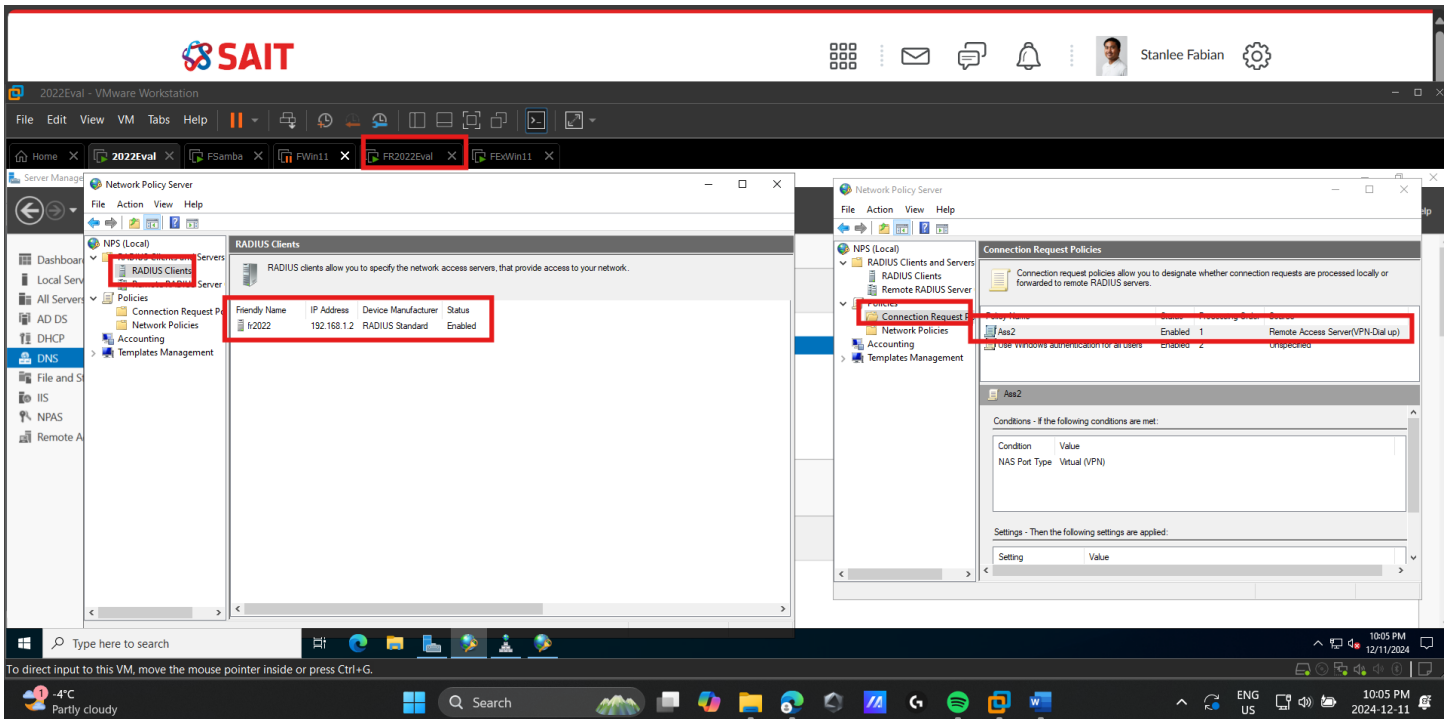
Screenshot of ADUC contents with users and groups from script



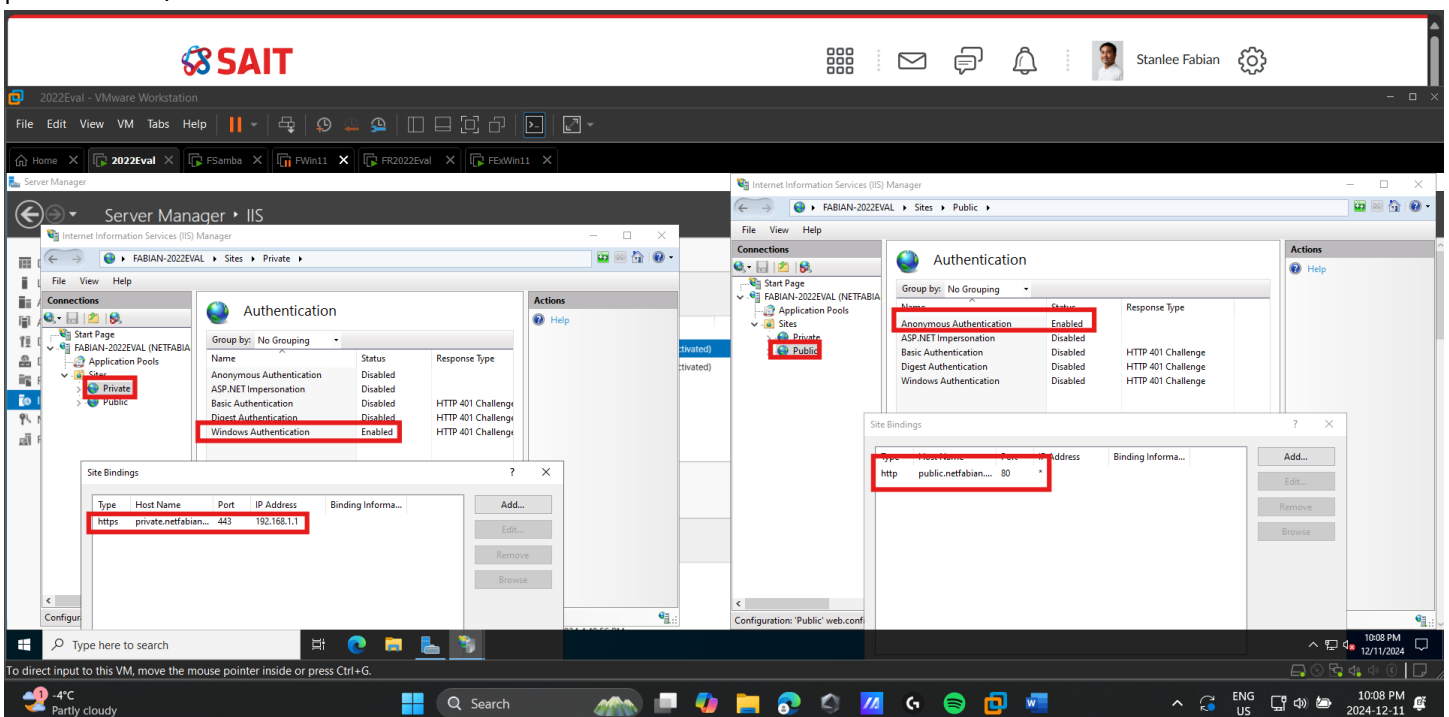
Contents of the DNS forward lookup zone with all records showing



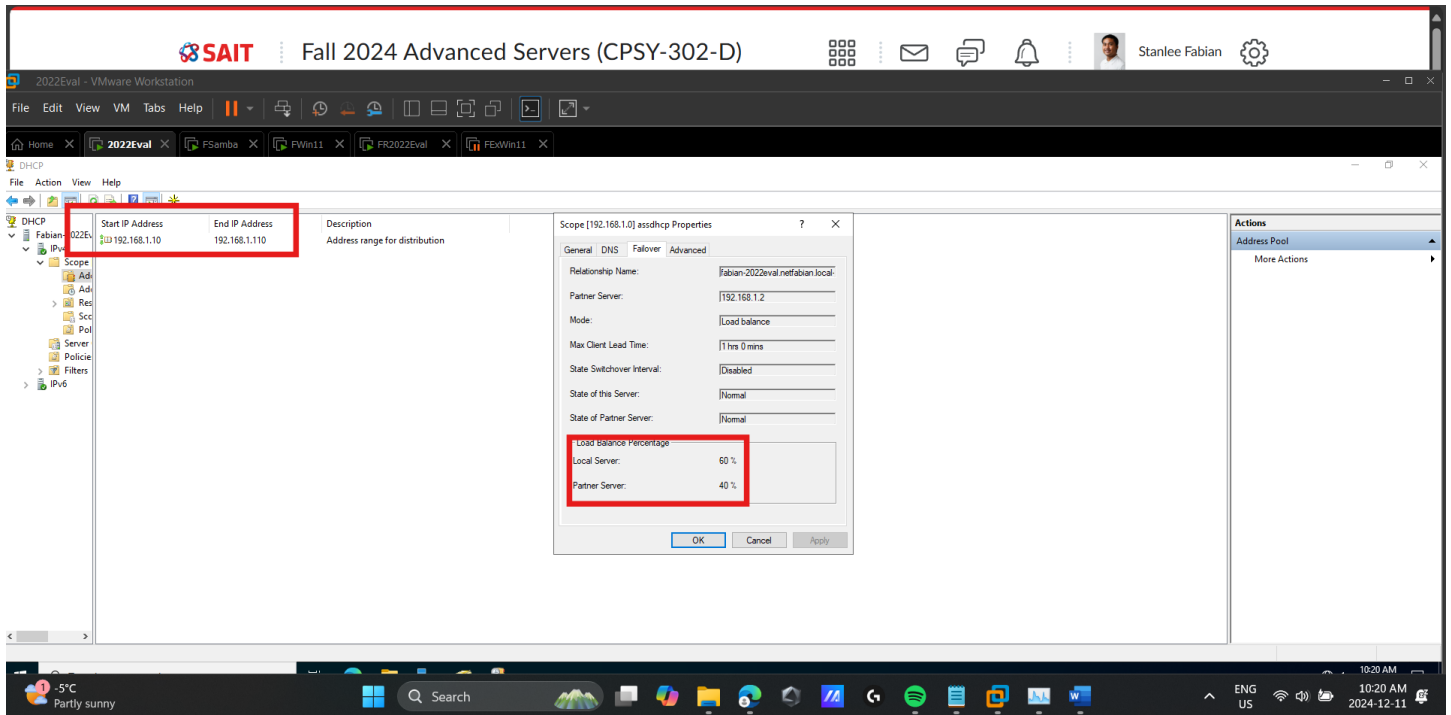
NPS configuration showing the contents of the radius client and the connection requests policy.



IIS configure showing configuration for each Website including the bindings and how permission/authentication is set



DHCP configuration showing scope and failover configuration – (IPv4 > scope > properties > failover tab)



(Share Perms and NTFS) Screenshot showing HOW permissions have been set on the "sales" share

SAIT Fall 2024 Advanced Servers (CPSY-302-D) Stanlee Fabian

2022Eval - VMware Workstation

File Edit View VM Tabs Help

Home x 2022Eval x FSamba x FWin11 x FR2022Eval x FExWin11 x

Active Directory Users and Computers

File Action View Help

File Home Share View

← → > This PC > Local Disk (C:) > Shares

Name Date modified Type Size

Quick access Desktop Downloads Documents Pictures This PC Desktop Documents Downloads Music Pictures Videos Local Disk (C:) Network

1 item 1 item selected

Type here to search

-7°C Cloudy

9:26 AM 12/11/2024

9:26 AM 2024-12-11

SALES PROPERTIES

Advanced Sharing

Share this folder

Settings

Share name: Sales

Limit the number of simultaneous users to: 16777

Comments:

Permissions Caching

OK Cancel Apply

PERMISSIONS FOR SALES

Share Permissions

Group or user names:

- Domain Admins (NETFABIAN\Domain Admins)
- Everyone
- Management (NETFABIAN\Management)
- Sales (NETFABIAN\Sales)

Permissions for Domain Admins

	Allow	Deny
Full Control	☒	☐
Change	☒	☐
Read	☒	☐

OK Cancel Apply

Select Users, Computers, Service Accounts, or Groups

Select this object type: Users, Groups, or Built-in security principals

From this location: netfabian.local

Enter the object names to select (examples): CREATOR OWNER

Check Names

Advanced... OK Cancel

SAIT Fall 2024 Advanced Servers (CPSY-302-D) Stanlee Fabian

2022Eval - VMware Workstation

File Edit View VM Tabs Help

Home x 2022Eval x FSamba x FWin11 x FR2022Eval x FExWin11 x

Active Directory Users and Computers

File Action View Help

File Home Share View

← → > This PC > Local Disk (C:) > Shares

Name Date modified Type Size

Quick access Desktop Downloads Documents Pictures This PC Desktop Documents Downloads Music Pictures Videos Local Disk (C:) Network

1 item 1 item selected

Type here to search

-8°C Mostly cloudy

9:33 AM 12/11/2024

9:33 AM 2024-12-11

SALES PROPERTIES

General Sharing Security Previous Versions Customize

Object name: C:\Shares\Sales

Group or user names:

- CREATOR OWNER
- SYSTEM
- Administrators (NETFABIAN\Administrators)
- Users (NETFABIAN\Users)

To change permissions, click Edit.

Permissions for Users

	Allow	Deny
Full control	☒	☐
Modify	☒	☐
Read & execute	☒	☐
List folder contents	☒	☐
Read	☒	☐
Write	☐	☐

For special permissions or advanced settings, click Advanced.

Close Cancel Apply

PERMISSIONS FOR SALES

Security

Object name: C:\Shares\Sales

Group or user names:

- Accounting (NETFABIAN\Accounting)
- CREATOR OWNER
- SYSTEM
- Administrators (NETFABIAN\Administrators)
- Management (NETFABIAN\Management)

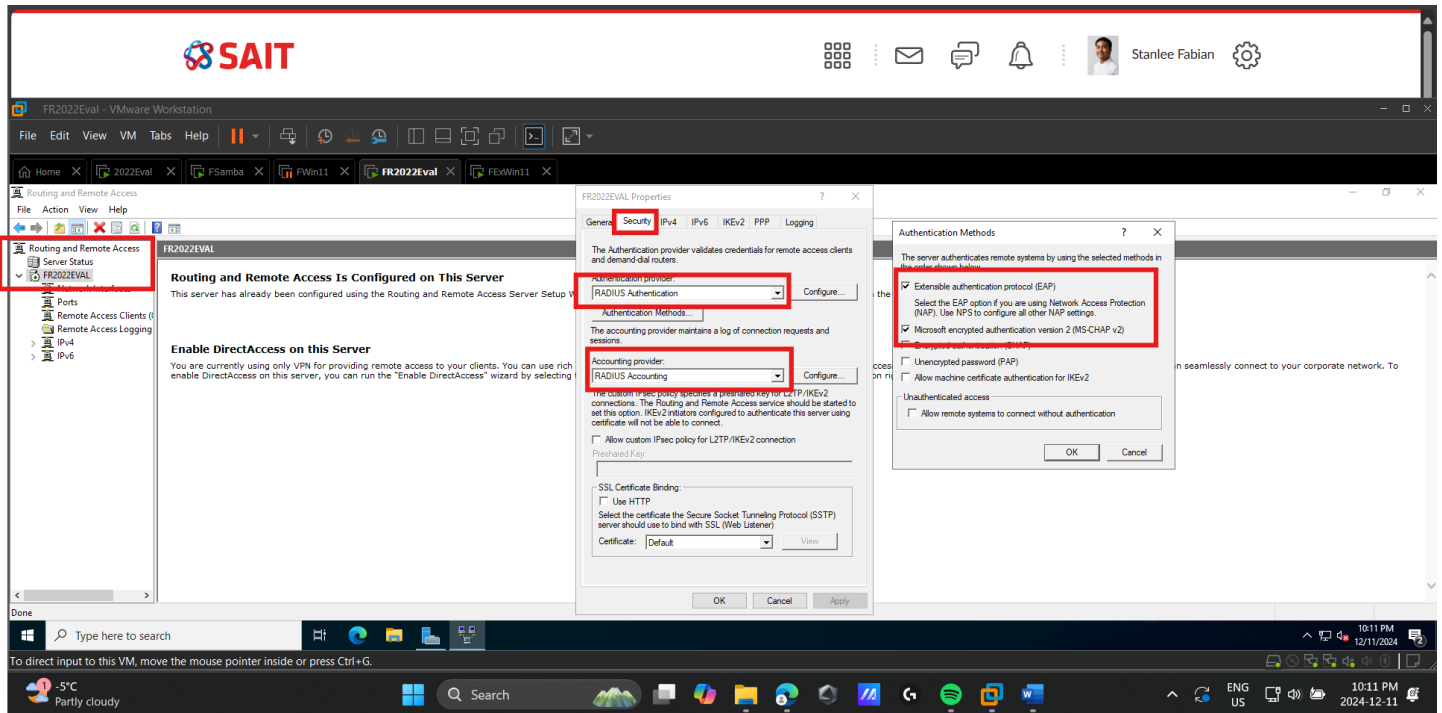
Permissions for CREATOR OWNER

	Allow	Deny
Full control	☒	☐
Modify	☒	☐
Read & execute	☒	☐
List folder contents	☒	☐
Read	☒	☐

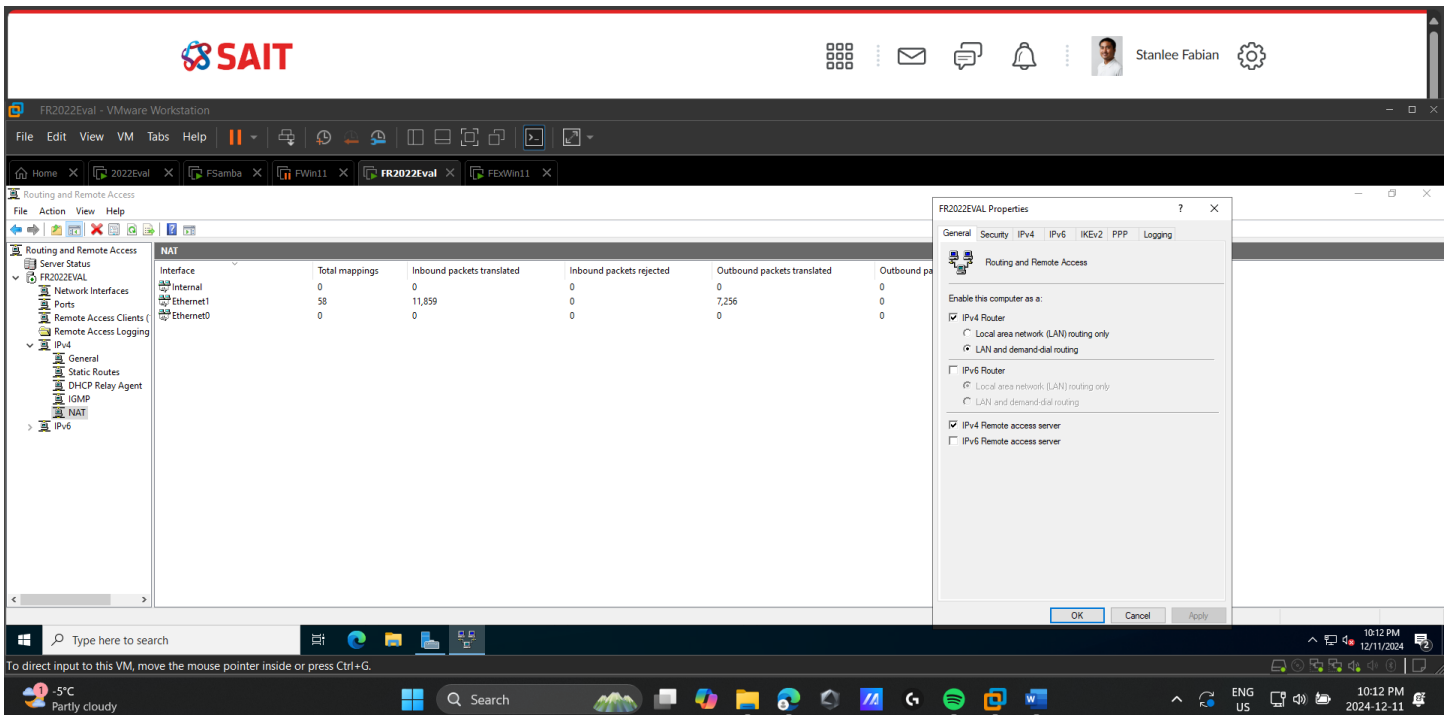
OK Cancel Apply

VPN Server

Security tab of RRAS properties, showing the configuration for RADIUS authentication and accounting



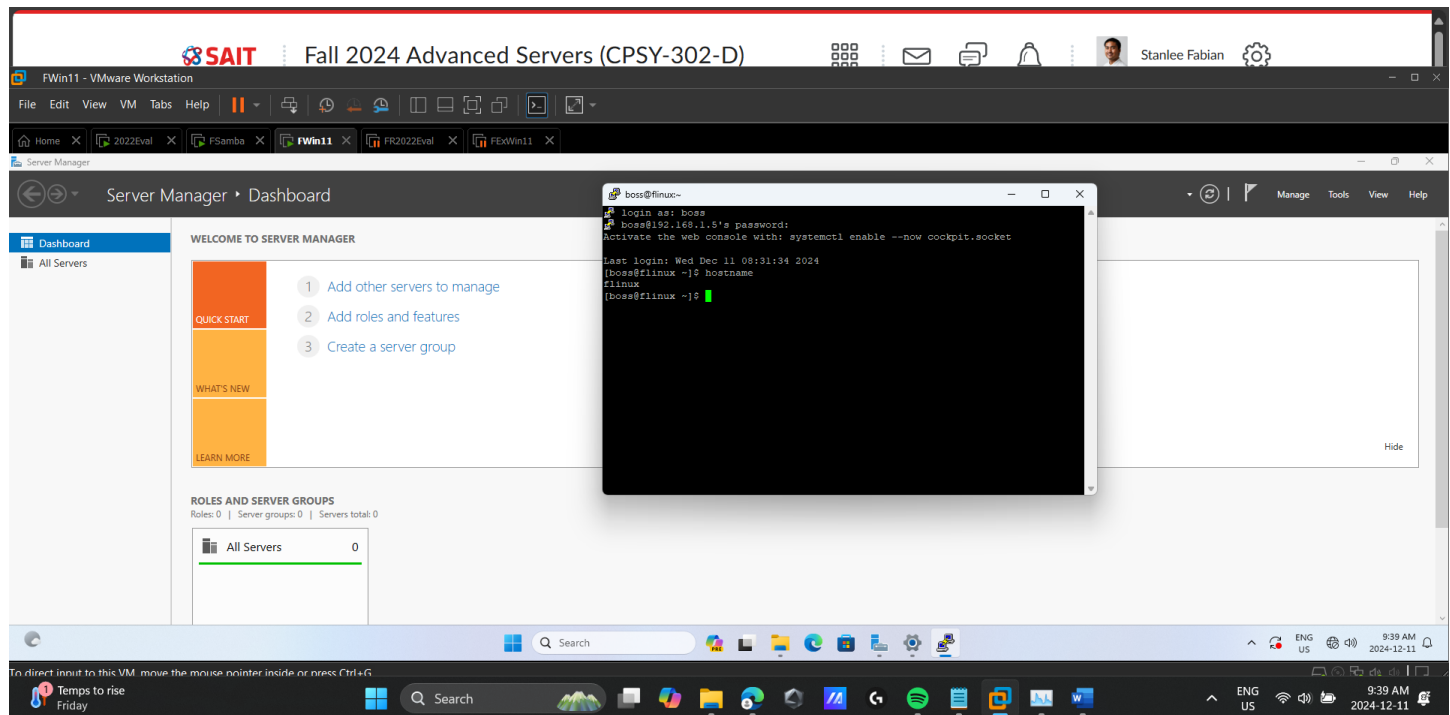
NAT config



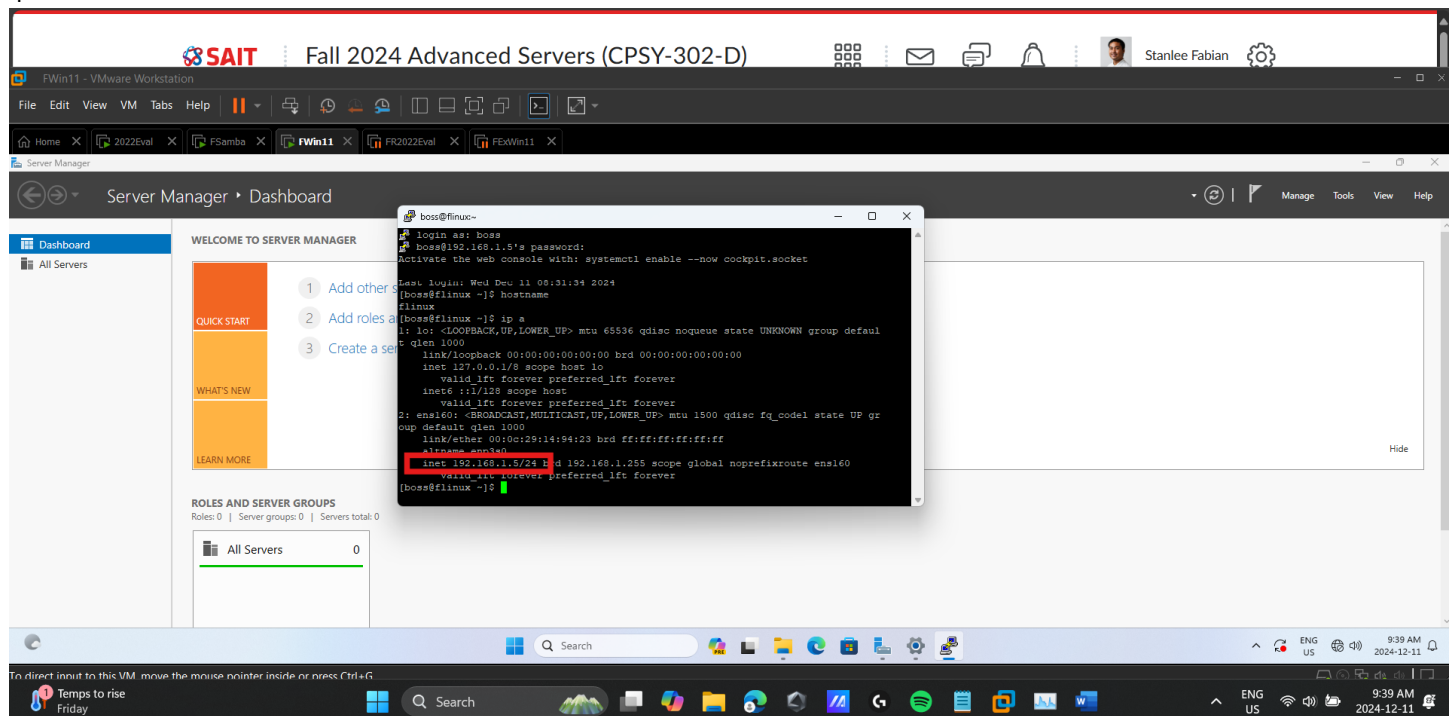
Linux server

Output of the following commands:

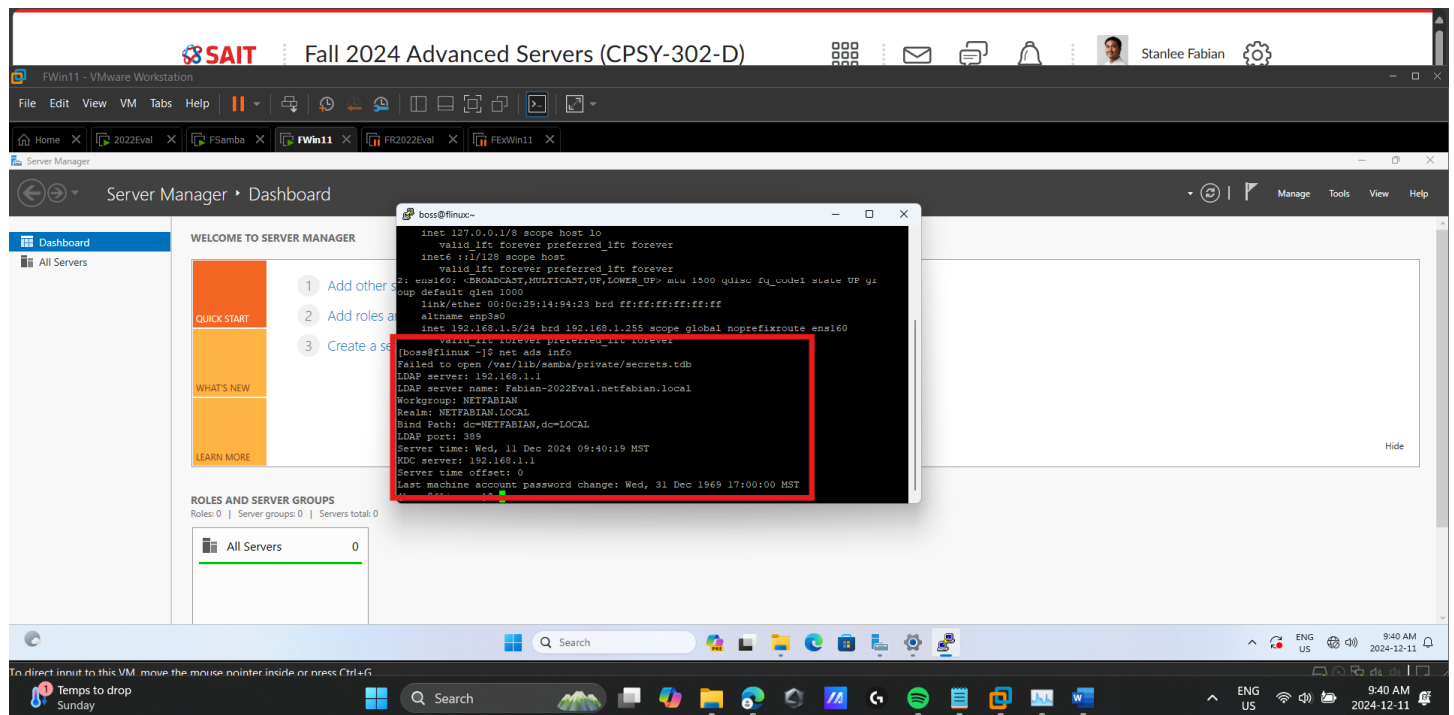
hostname



ip add



net ads info

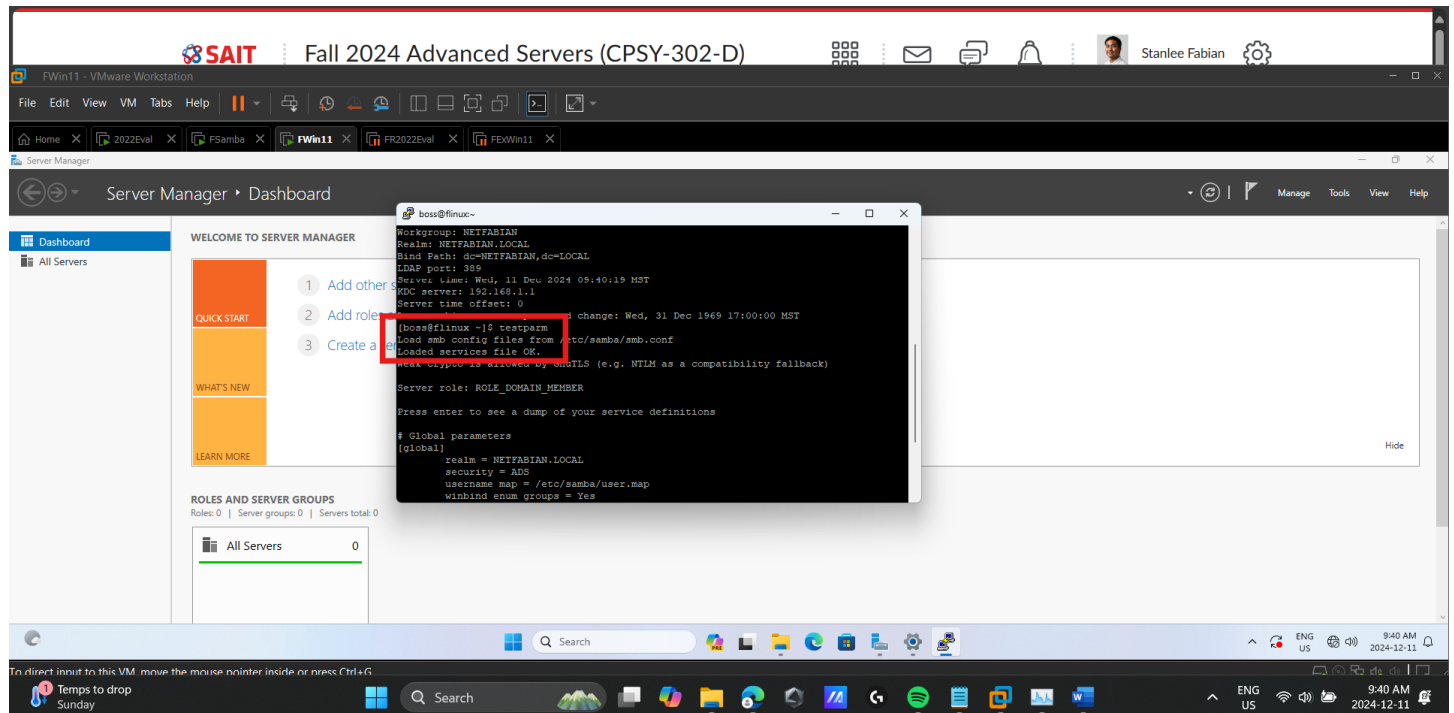


SAIT Fall 2024 Advanced Servers (CPSY-302-D)

Server Manager Dashboard

```
boss@finuc-
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qlen 1000 state UP ql
loop default qlen 1000
link/ether 00:0c:29:14:94:23 brd ff:ff:ff:ff:ff:ff
    altname ens380
    inet 192.168.1.5/24 brd 192.168.1.255 scope global noprefixroute ens160
        valid_lft forever preferred_lft forever
[boss@finuc ~]$ net ads info
Failed to open /var/lib/samba/private/secrets.tdb
LDAP server: 192.168.1.1
LDAP server name: Fabian-2022Eval.netfabian.local
Workgroup: NETFABIAN
Realm: NETFABIAN.LOCAL
Bind Path: dc=NETFABIAN,dc=LOCAL
LDAP port: 389
Server time: Wed, 11 Dec 2024 09:40:19 MST
KDC server: 192.168.1.1
Server time offset: 0
Last machine account password change: Wed, 31 Dec 1969 17:00:00 MST
```

testparm



SAIT Fall 2024 Advanced Servers (CPSY-302-D)

Server Manager Dashboard

```
boss@finuc-
Workgroup: NETFABIAN
Realm: NETFABIAN.LOCAL
Bind Path: dc=NETFABIAN,dc=LOCAL
LDAP port: 389
Server time: Wed, 11 Dec 2024 09:40:19 MST
KDC server: 192.168.1.1
Server time offset: 0
Last machine account password change: Wed, 31 Dec 1969 17:00:00 MST
[boss@finuc ~]$ testparm
Load smb config files from /etc/samba/smb.conf
Loaded services file OK.
Weak crypto is allowed by default (e.g. NTLM as a compatibility fallback)
Server role: ROLE_DOMAIN_MEMBER
Press enter to see a dump of your service definitions
# Global parameters
[global]
    realm = NETFABIAN.LOCAL
    security = ADS
    username map = /etc/samba/user.map
    winbind enum groups = Yes
```

cat /etc/samba/smb.conf

The screenshot shows a VMware Workstation interface with a Linux virtual machine named 'FWin11'. A terminal window is open, displaying the output of the command 'cat /etc/samba/smb.conf'. The configuration file content is as follows:

```
[accounting]
guest ok = Yes
path = /shares/domain/accounting
write list = @NETFABIAN\accounting

[global]
workgroup = NETFABIAN
netbios name = flinux
security = ads
realm = NETFABIAN.LOCAL
idmap config * : backend = autorid
idmap config * : range = 1000000-1999999
winbind enum users = Yes
winbind enum groups = Yes
username map = /etc/samba/user.map

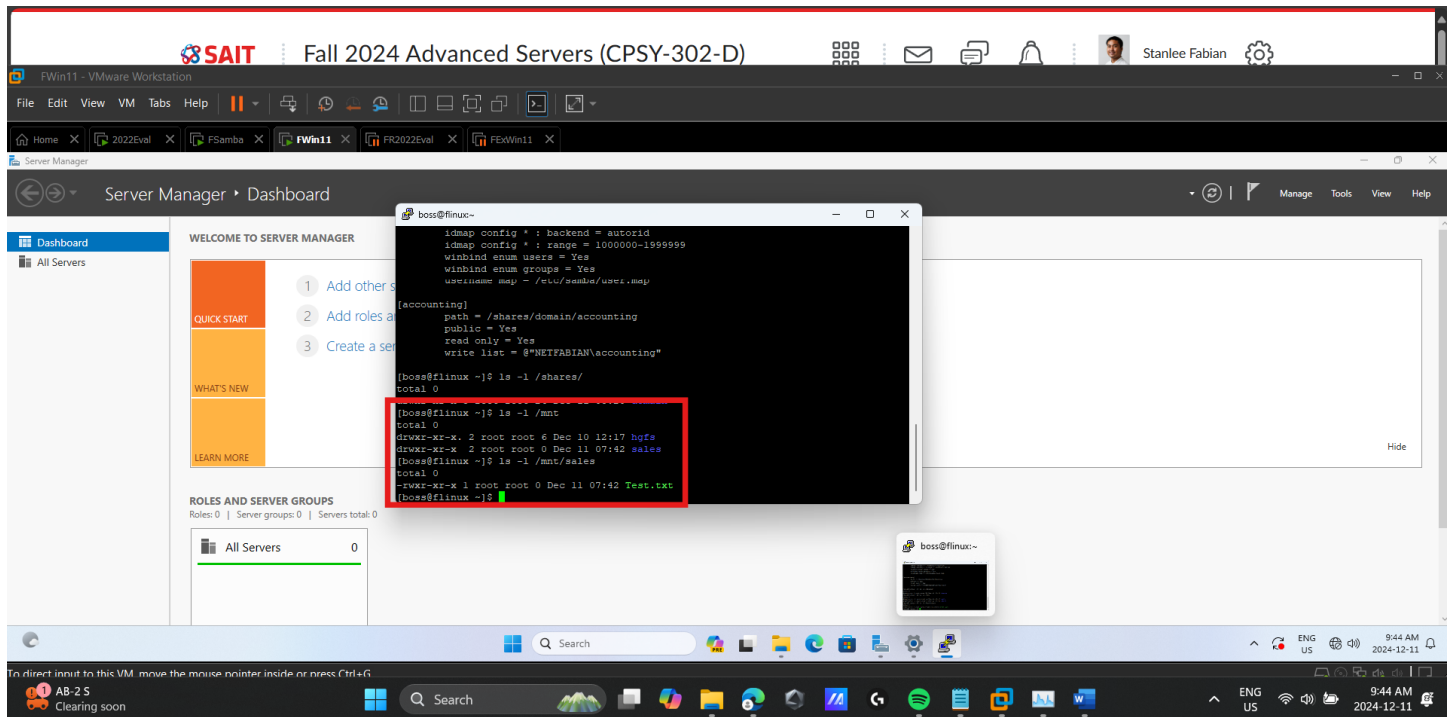
[accounting]
path = /shares/domain/accounting
public = Yes
read only = Yes
write list = @"NETFABIAN\accounting"
```

ls -l /shares/

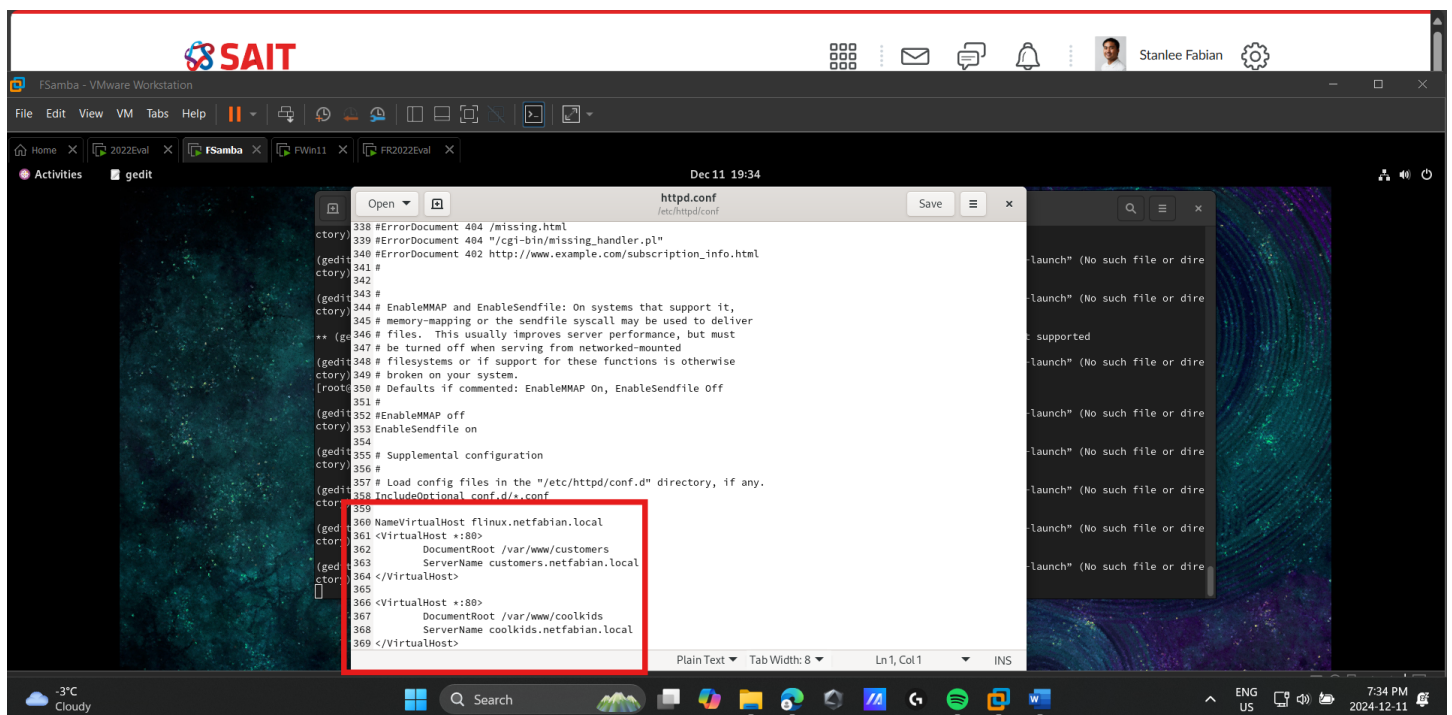
The screenshot shows the same VMware Workstation interface. The terminal window now displays the output of the command 'ls -l /shares/'. The output is as follows:

```
total 0
drwxr-xr-x 3 root root 24 Dec 11 08:20 domain
```


Mounted Sales Share from ADC



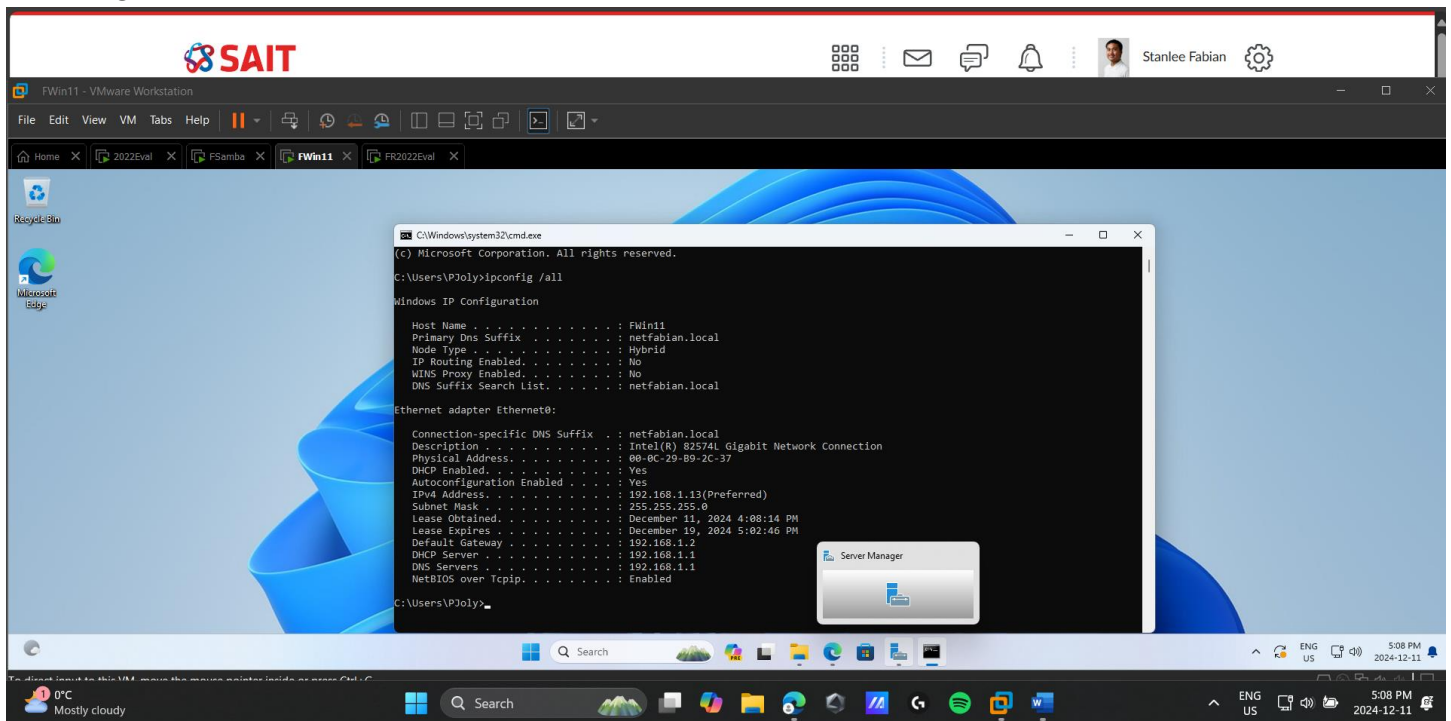
Screenshot from the httpd configuration displaying the settings for sharing the two websites as virtual directories



Internal Client

==When logged in as Pascal Joly==

ipconfig /all output



```
C:\Windows\system32\cmd.exe
(c) Microsoft Corporation. All rights reserved.

C:\Users\PJoly>ipconfig /all

Windows IP Configuration

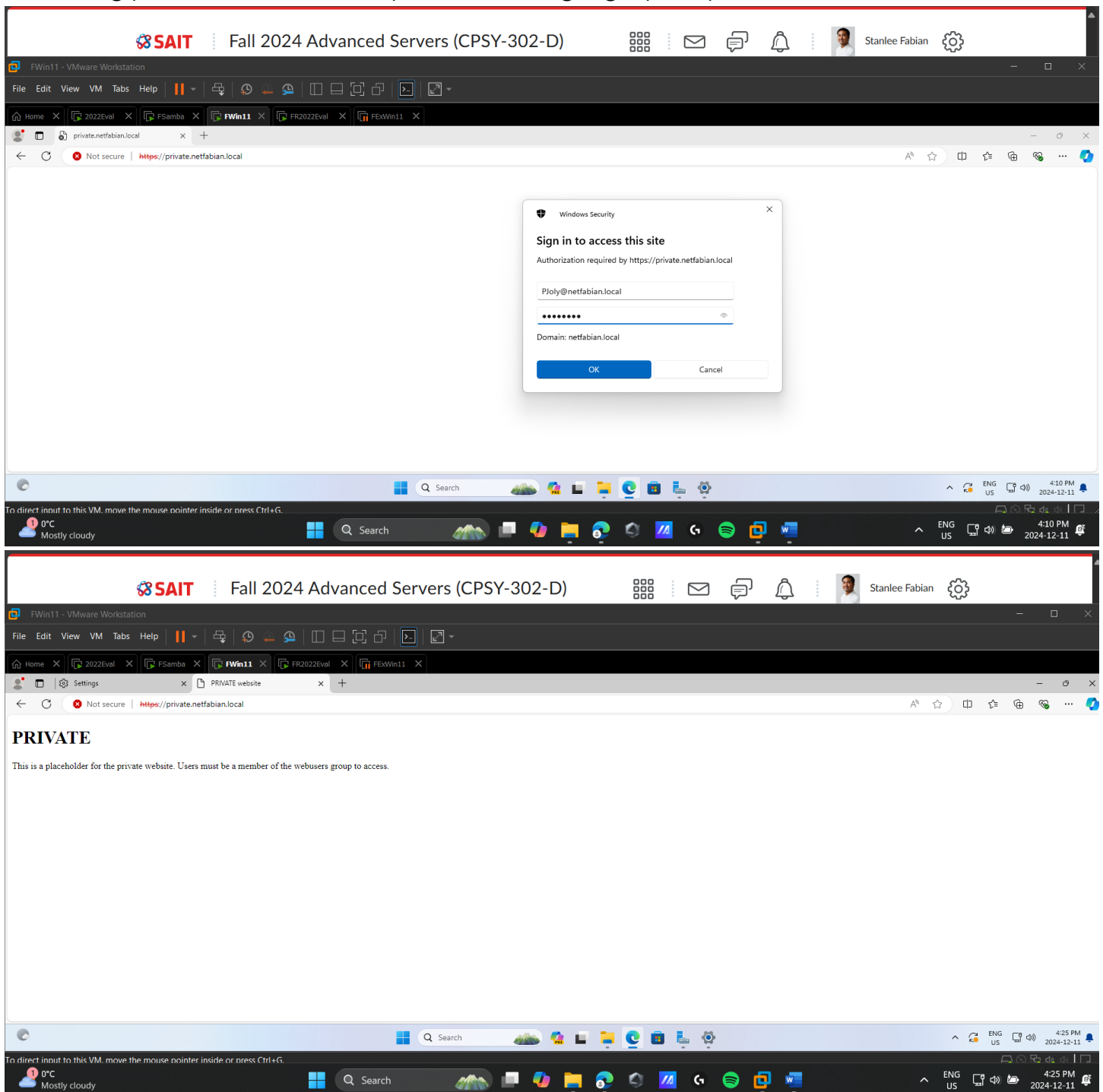
Host Name . . . . . : FWin11
Primary Dns Suffix . . . . . : netfabian.local
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : netfabian.local

Ethernet adapter Ethernet0:

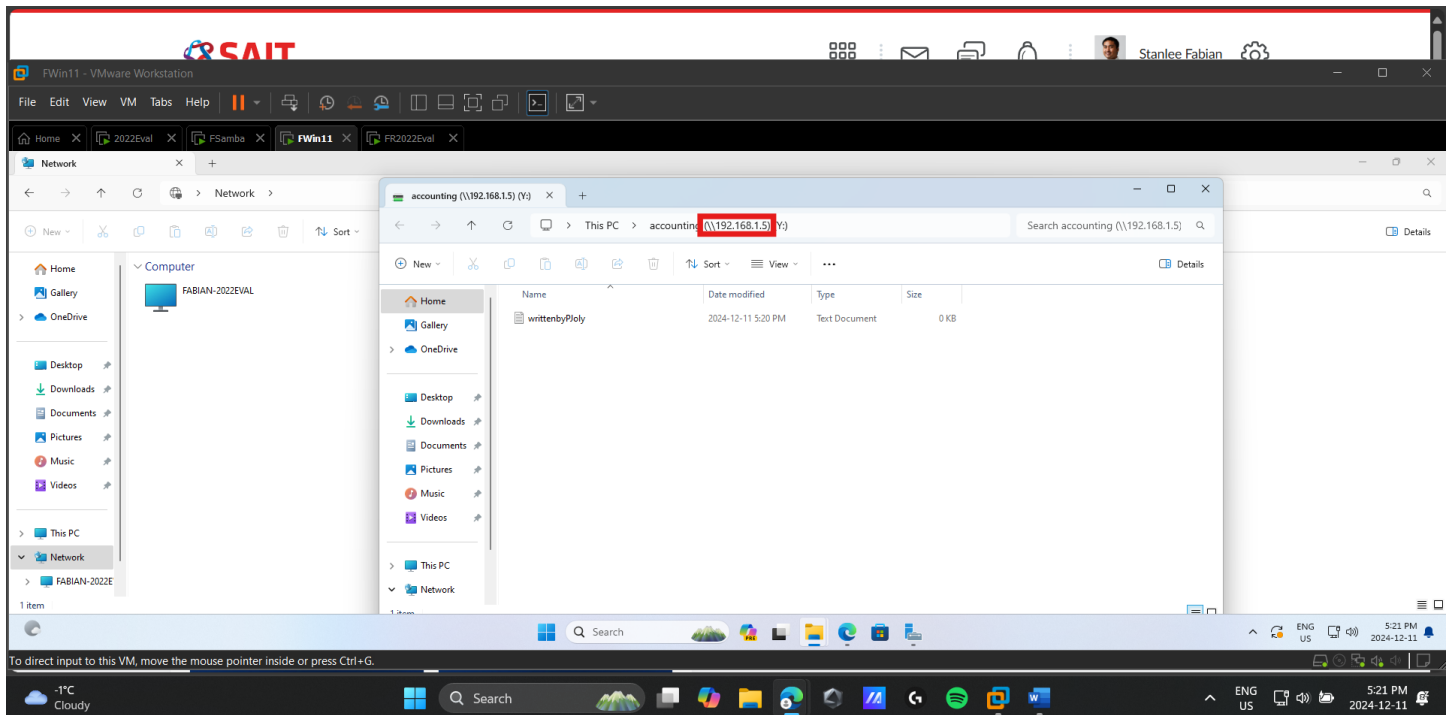
Connection-specific DNS Suffix . : netfabian.local
Description . . . . . : Intel(R) 82574L Gigabit Network Connection
Physical Address. . . . . : 08-0C-29-89-2C-37
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 192.168.1.13(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : December 11, 2024 4:00:14 PM
Lease Expires . . . . . : December 19, 2024 5:02:46 PM
Default Gateway . . . . . : 192.168.1.2
DHCP Server . . . . . : 192.168.1.1
DNS Servers . . . . . : 192.168.1.1
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\PJoly>
```

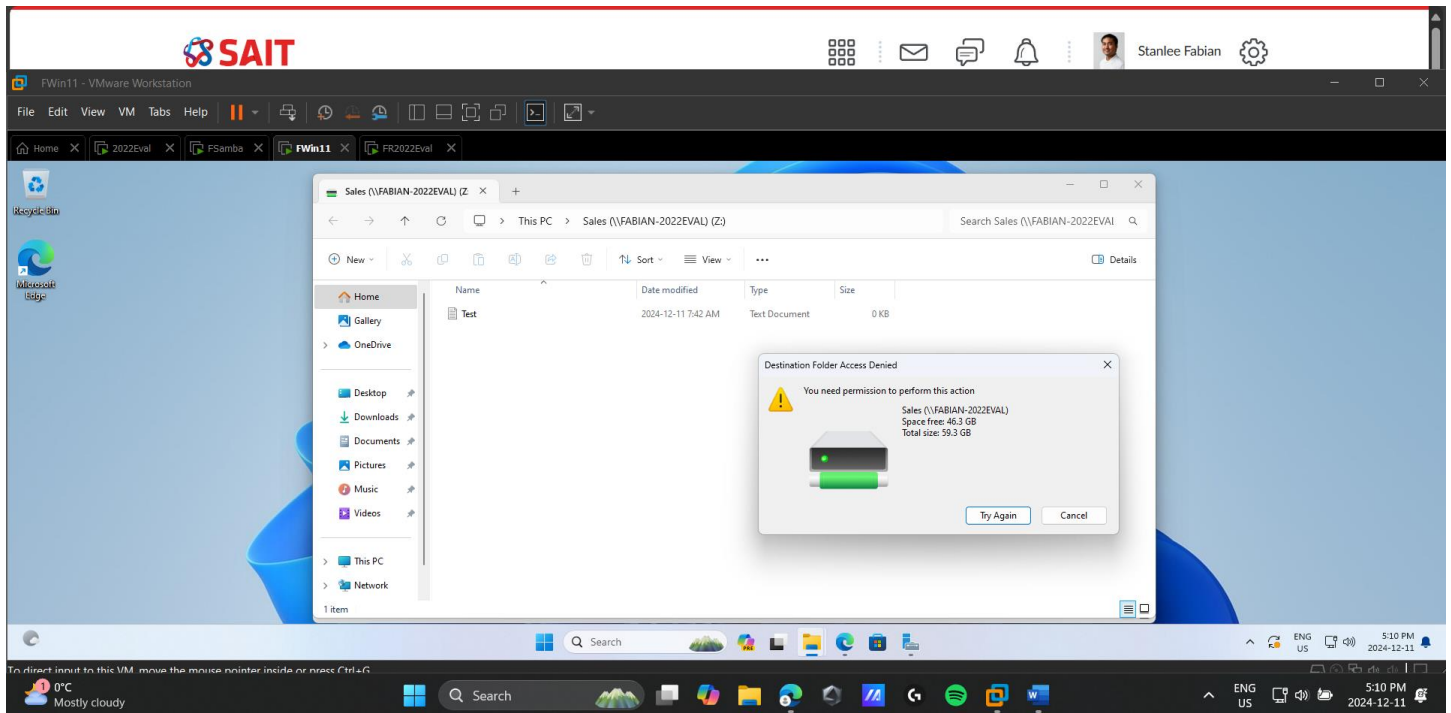
Accessing private website via https:// - including login prompt



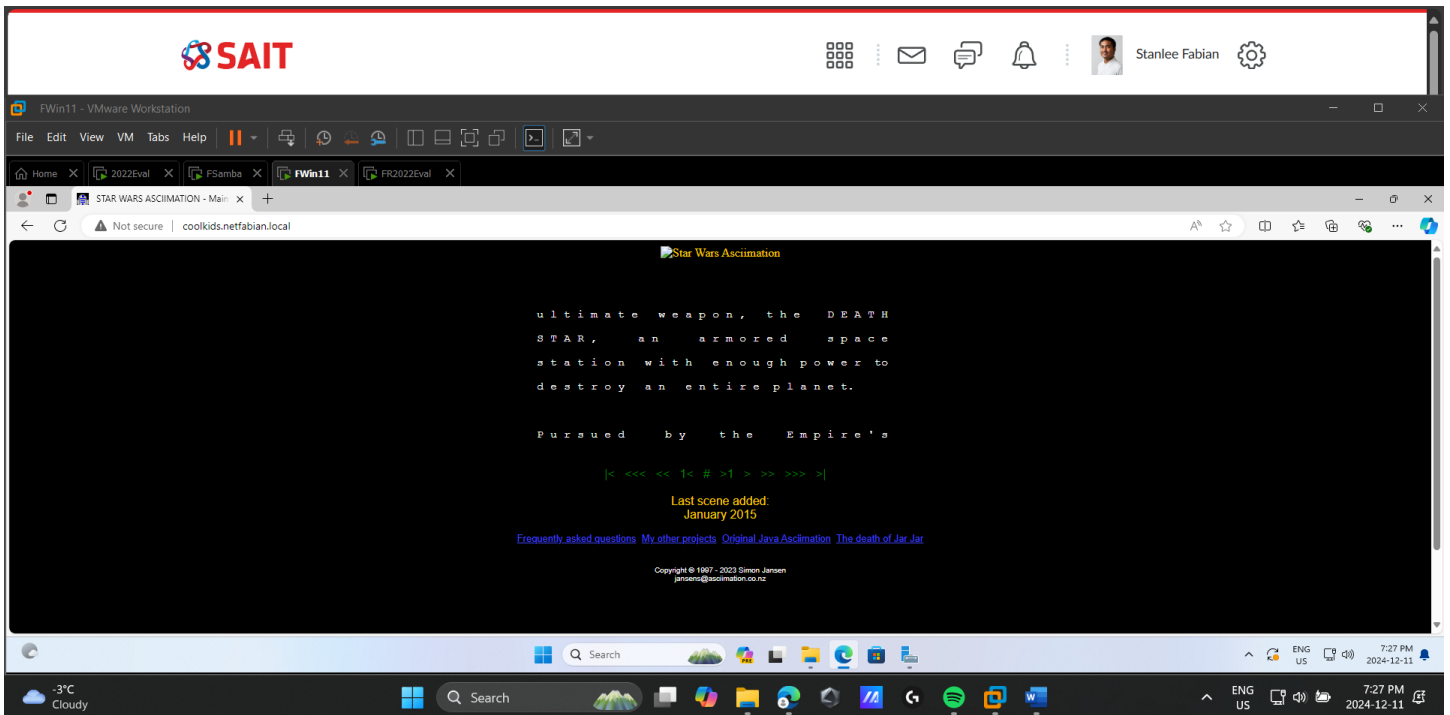
Accessing and being able to write to "accounting" share on Linux machine



Unable to read but not write to "sales" share on DC

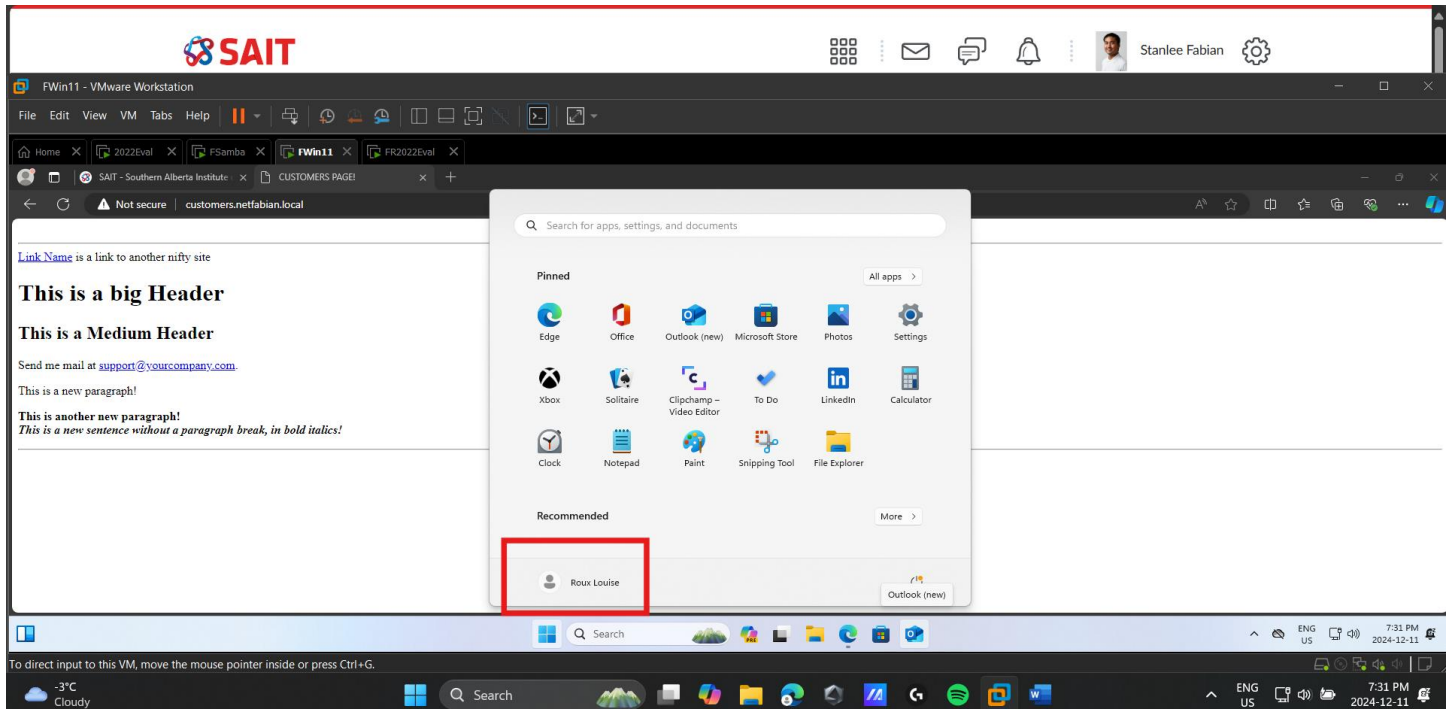


Accessing coolkids website via http://

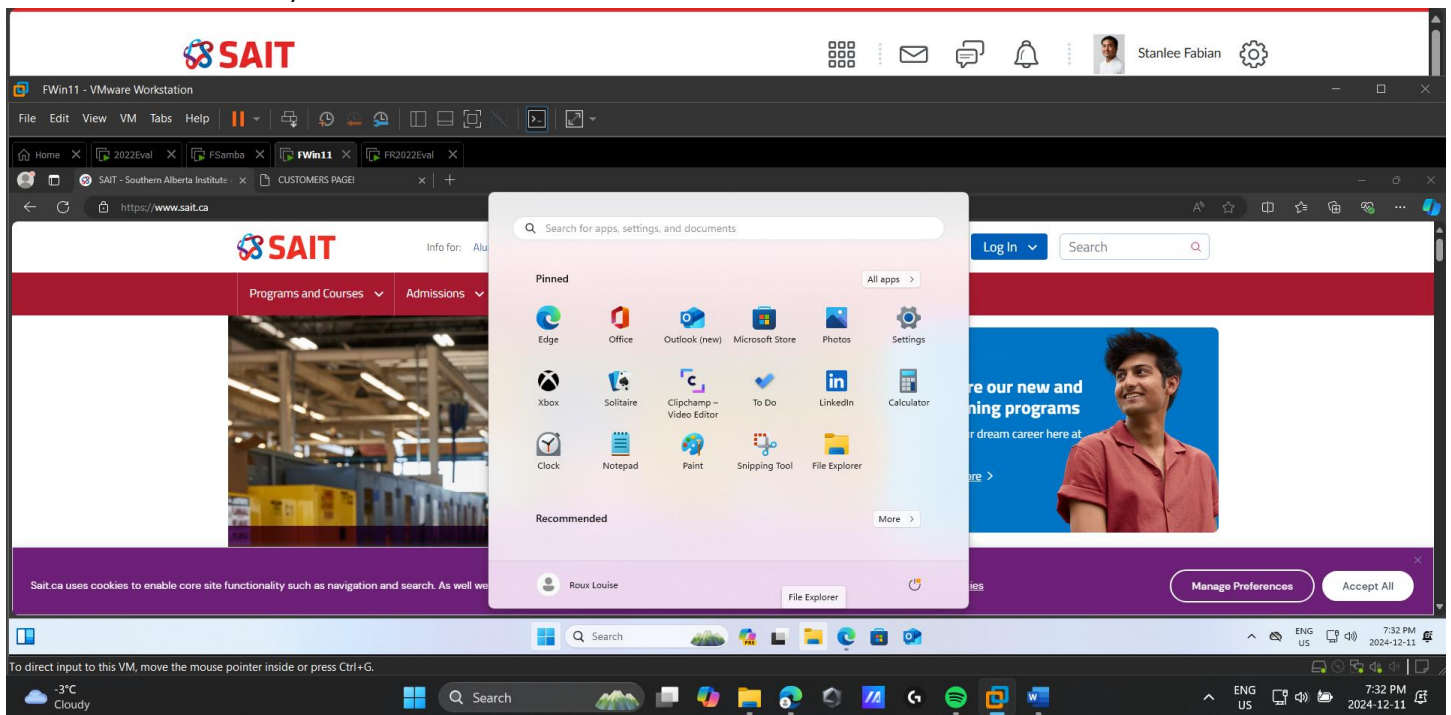


==When logged in as Louise Roux==

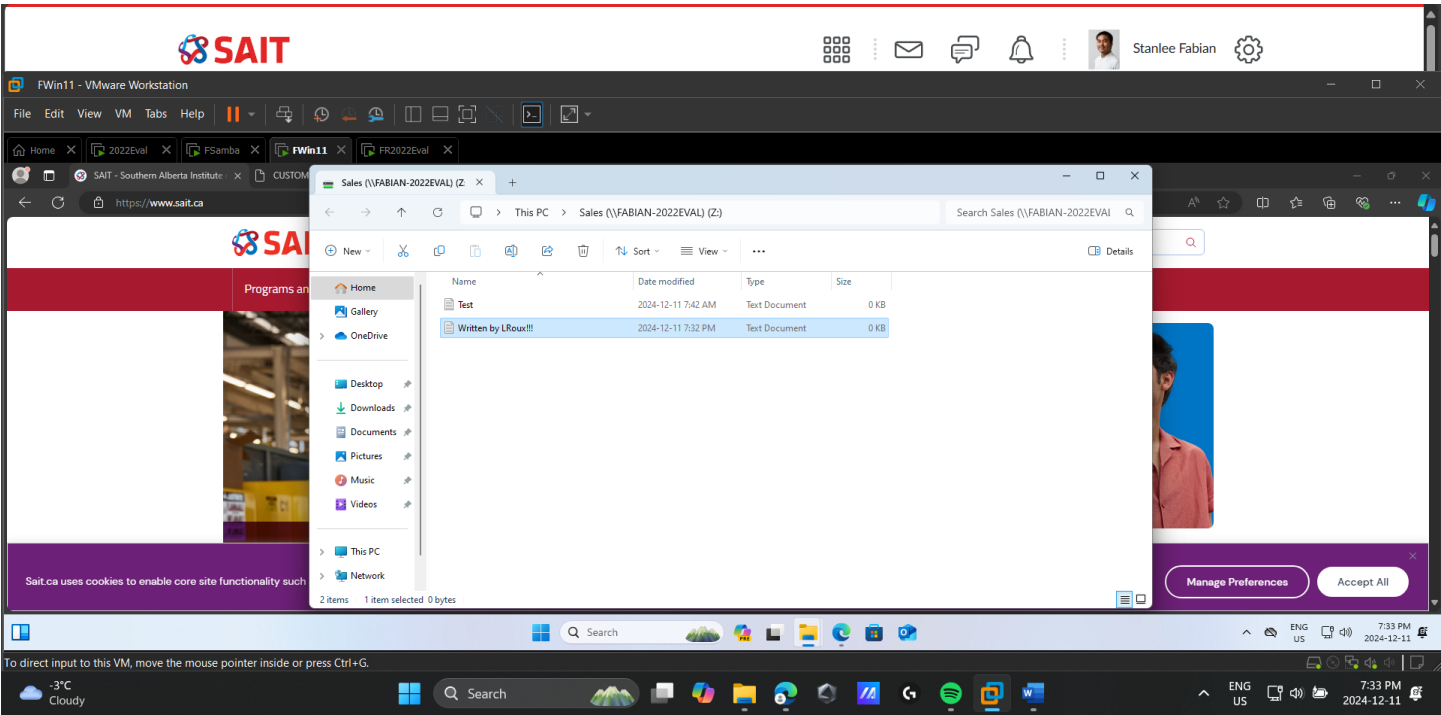
Able to access and using <http://customers.mydomain.com>



Able to access any Internet site with browser



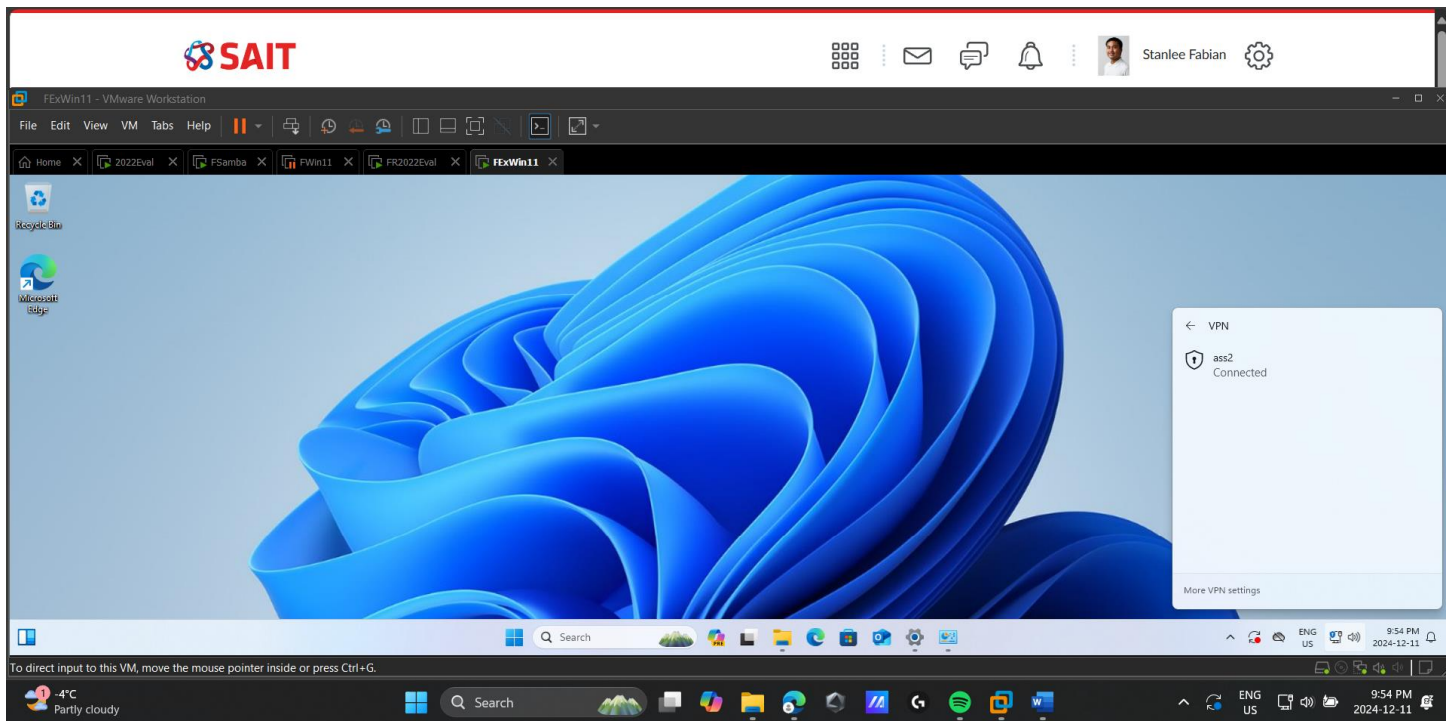
Able to write to "sales" share



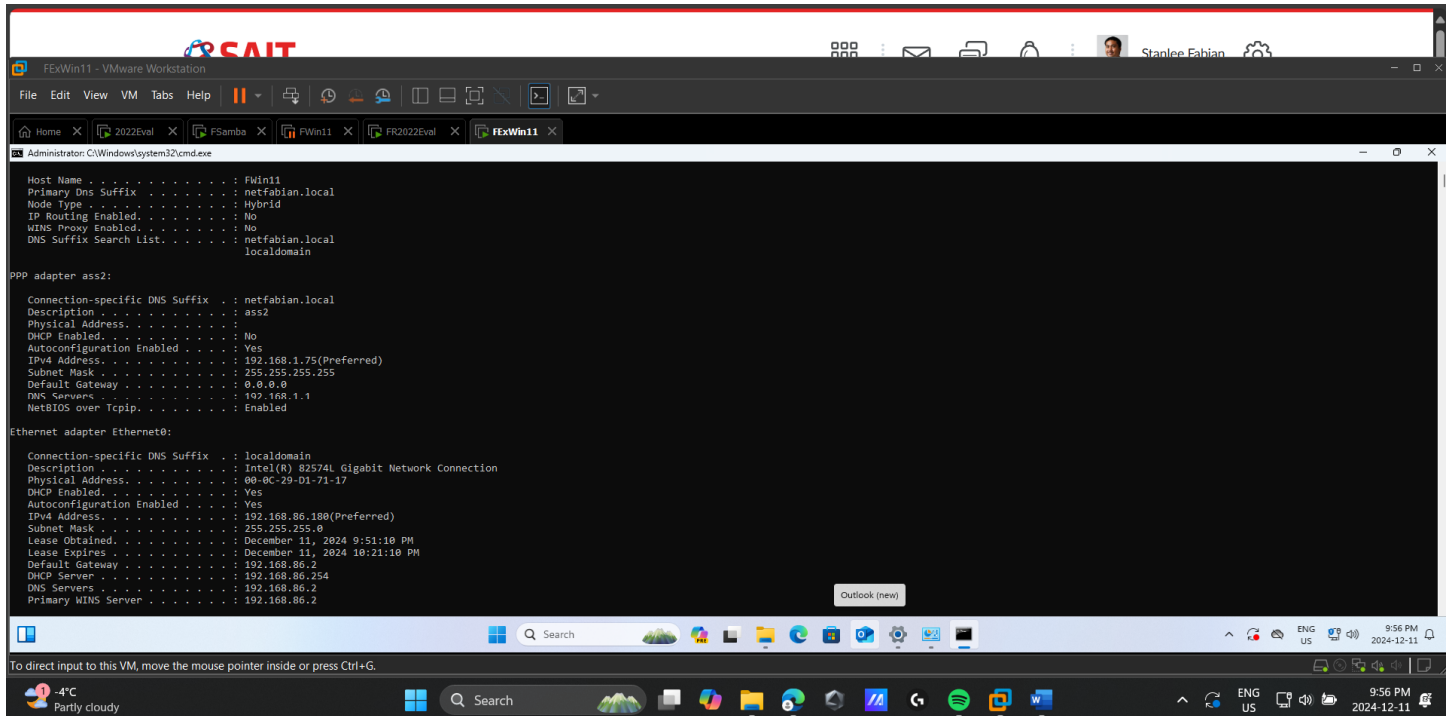
External client

==When logged in as meadmin account==

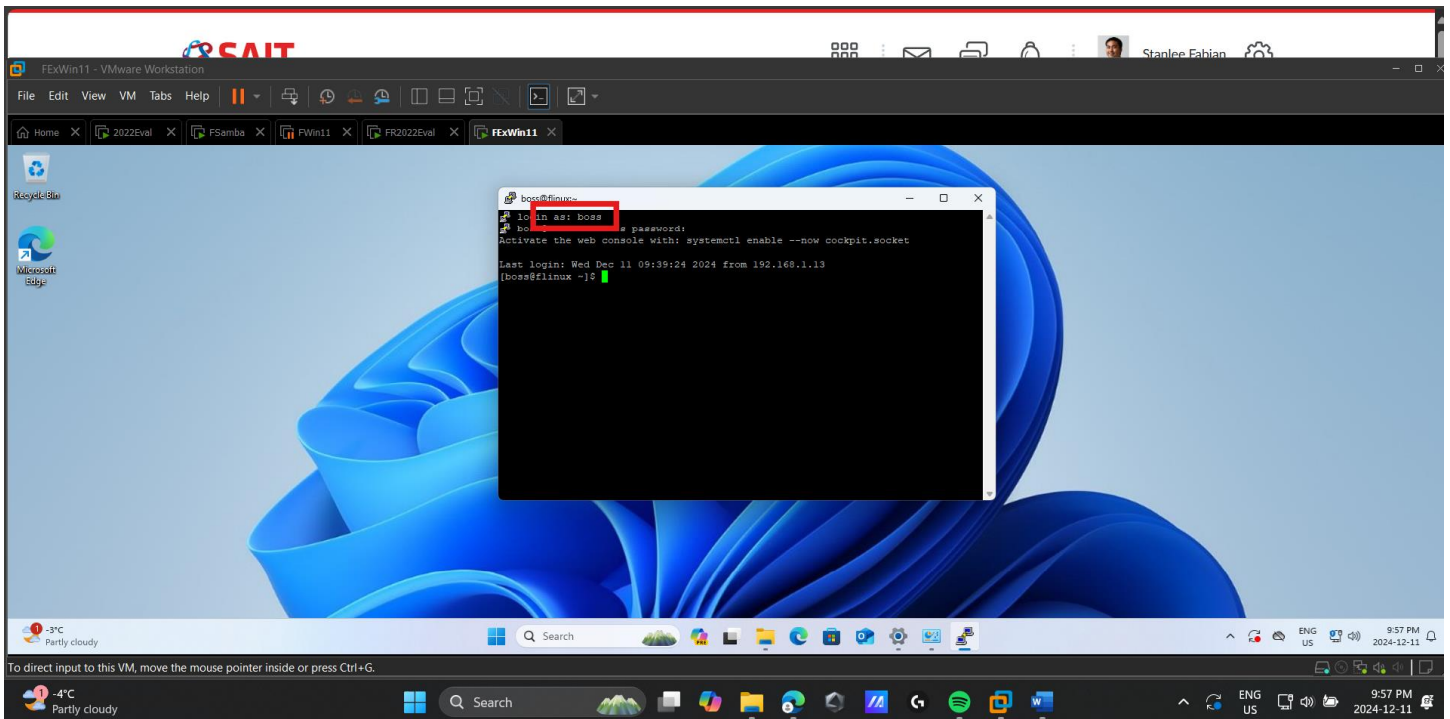
Successful VPN connection



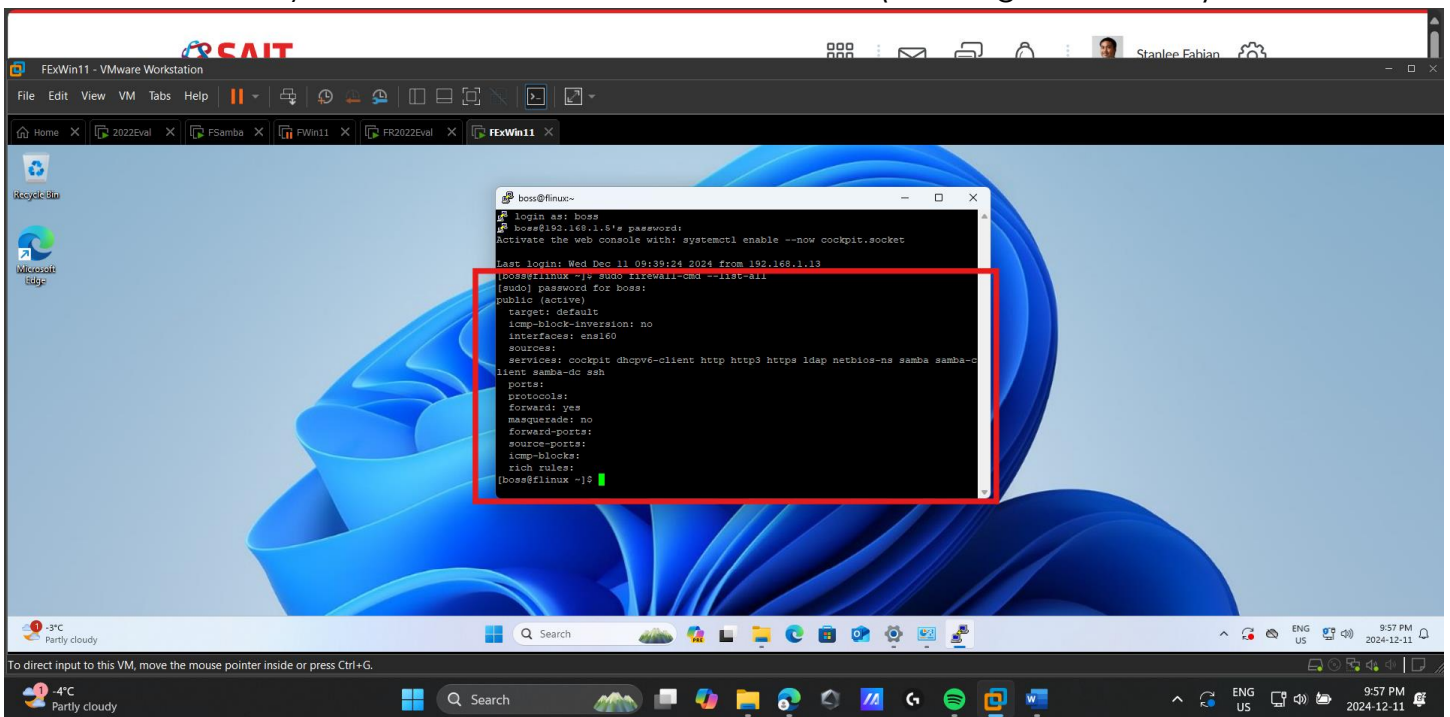
Ipconfig /all when VPN connected



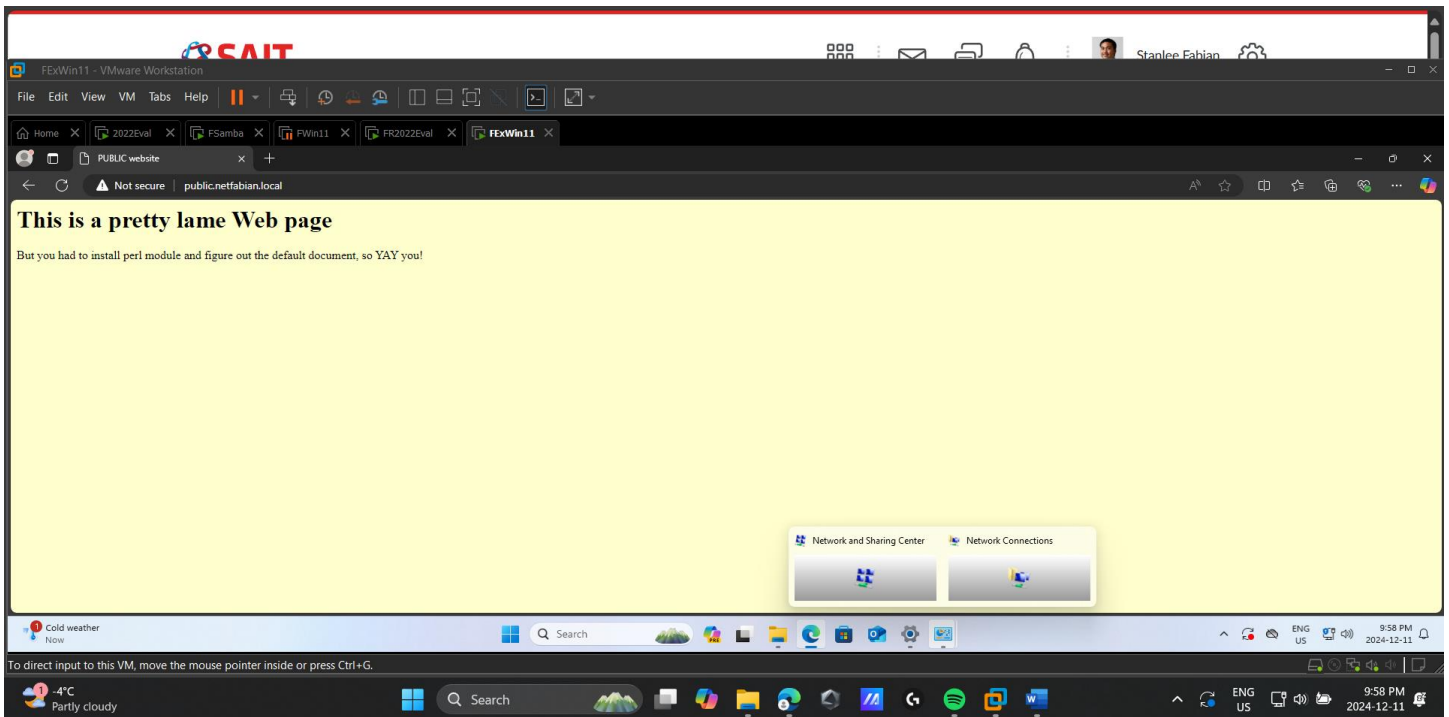
Successful ssh to linux machine as "boss" user



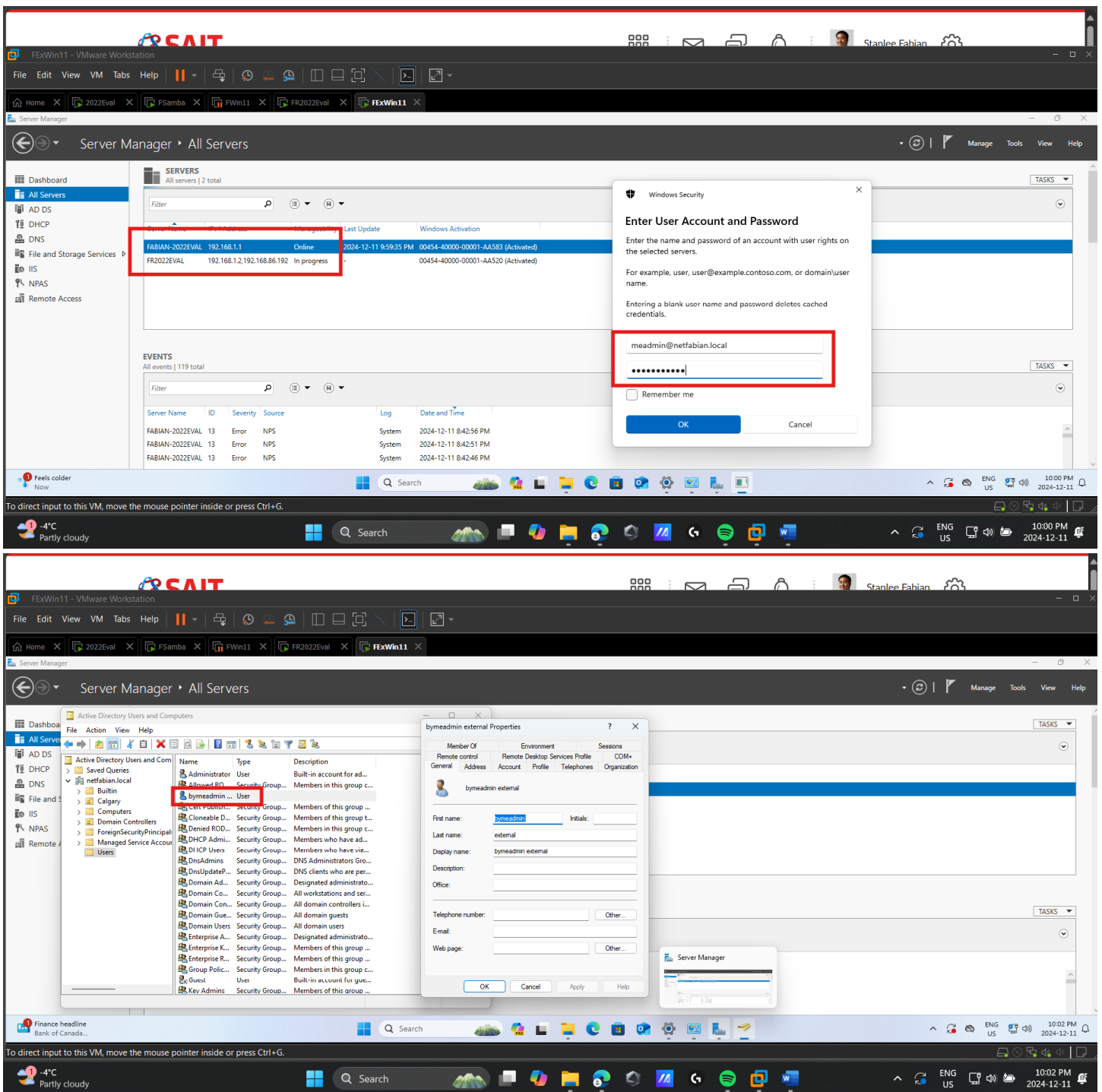
Able to successfully execute "sudo firewall-cmd --list-all" (showing use of sudo) as boss user



Accessing public site via http://



Show use of server manager with both servers added to console and managed as meadmin account, creating a user in AD



Platform Comparison

Remember to include your paragraph with your platform preference

I consider myself a system administrator noob, but in all honesty, I like Linux! It provides so many options for customization – text editors alone are multiple. I prefer that it doesn't push or make you do things a certain way. It is also much lighter in memory, could be built from the ground up to your liking, and the best part, open source! Hence the many flavors.

I have also had experience with hackathons using Linux flavors and I feel like I just associated the feeling I get from those events to the system but nonetheless a valid reason.